CANALRUNG · PROJECT PLAN

CanalRung project plan

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DOCUMENT CONTROL

Document ID	CNR-PLN-001	Version	0.1
Status	Draft	Date	2026-11-05
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/11-plan.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First project plan (kit 1.9.1 upgrade): stage P3 reached; P4 to P6 on hold under the TRL 3 cap

CanalRung project plan

CanalRung is at stage P3 (design outputs, TRL 3) for option A, with options B and C modelled at the same detail; the next gate is Amish's choice among the options (CNR-DDR-004, proposed, awaiting Amish), after which the stage cap decides whether verification (P4) may start. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem, concept and requirements	Design lead	CNR-PRB-001, CNR-PRC-001, CNR-URS-001, CNR-PRS-001	Done (P3)
STEM check and concept options	Design lead	CNR-STM-001, CNR-OPT-001, CNR-DDR-004	Done; choice proposed, awaiting Amish; independent review recorded
Calculations	Design lead	CNR-CAL-001 (option A only); options B and C screened in the STEM check	Done for A
Detailed CAD	Design lead	`cad/src/model.py` for A, B and C at D3; STEP, drawings, media	Done
Pipeline documents	Design lead	CNR-REG-001, CNR-FEA-001, CNR-RSK-001, CNR-MKT-001, CNR-PLN-001, CNR-RTM-001	Done (first issue)
Build plan and BOM	Design lead	CNR-BLD-001, `bom/bom.csv`	Done (option A)
Partner contact	Amish	Kittitas Reclamation District; Water Safety Council of Fresno County	Not started; not contacted
Verification (P4)	To be named	Proof loads, throw and climb trials, UV ageing	On hold

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead (Claude with Amish's review)	P0 to P3 documents	Done	0
Prototype parts	One option A station (P4, on hold)		605.20 (BOM, indicative; Amish decides)
Proof-load rig (CalRig, a candidate shared block)	Rope, rung and anchor proof loads (P4, on hold)	To assess	To assess
Supervised test tank and lifeguards	Climb and throw trials (P4, on hold)	To assess	To assess
Partner site survey	Heights, slopes, soils, flows	To assess	To assess
Regulatory reviewer	Classification and approval route (CNR-REG-001 G3)	To assess	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done 2026-11-05 (kit 1.9.1 documents)
P1 Definition	URS, PRS draft, RSK, MKT, REG	Done 2026-11-05
P2 Design inputs	PRS targets, CAL, RTM, DDR, STEM check closed	Done 2026-11-05; twelve STEM rows stay open for tests, datasheets or Amish's decisions (S4, S5, S8, S9, S10, S11, S12, S15, S17, S18, S22, S23)
P3 Design outputs	Model, STEP, DWG, BOM, BLD, DEC, FMEA	Done 2026-11-05
P4 Verification	Proof loads, throw and climb trials, UV ageing	On hold
P5 to P6	Field trials with a partner, launch	On hold

Next steps and decision requested

Decision requested from Amish: choose among options A, B and C (recommended: A, CNR-DDR-004), and decide the three fixes proposed after the independent review (CNR-DEC-001, open decisions 3 to 5: a deeper footing, the rope margin rule, the eye nut off axis). Also open: whether to keep the USD 605.20 cost against the USD 500 target as a reported overrun (the target is not changed here), and whether to approve contacting the two candidate partners. After the cap is lifted: proof loads of the rope and a roped rung, a supervised tank trial of climbing and throwing, and an ageing test (all TRL 4 and not started).

References

- CNR-PRB-001 prior work table, for the competitor sources.
- See the source lists in CNR-STM-001, CNR-REG-001 and CNR-MKT-001.